

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:50

Annexure A

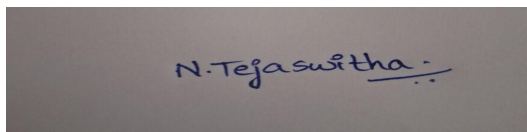
Technical Presentation

Criteria	Technical Content Accuracy(3)	Problem Identification(2.5)	Use of Tools(2)	Communication(1.5)	Professionalism(1)	Total(10)
Weightage	30%	25%	20%	20%	10%	100%
Q1						
Q2						

Code of Conduct:

I _____N.Tejaswitha 192371057_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

Focus Areas:

- Definition and need of DSS
 - Components (Data, Model, UI)
 - Types of DSS
 - Real-world applications (banking, healthcare)
 - Advantages & limitations
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Topic 2: Operational Systems vs DSS

Focus Areas:

1. OLTP vs OLAP concepts
 2. Key differences (data, users, purpose)
 3. Performance comparison
 4. Use-case scenarios
 5. Integration between systems
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Topic 3: Architecture of Data Warehouse

Focus Areas:

1. Data warehouse definition
 2. Three-tier architecture
 3. ETL process
 4. Data marts & metadata
 5. Benefits in decision-making
-

Topic 4: OLAP and Data Cubes

Focus Areas:

1. OLAP concept
 2. Data cube structure
 3. OLAP operations (slice, dice, roll-up, drill-down)
 4. Applications in analytics
-

Topic 5: Dimensional Data Modeling

Focus Areas:

1. Fact and dimension tables
 2. Star schema vs snowflake schema
 3. Schema design process
 4. Real-world example
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Topic 6: Query Processing & BI Tools

Focus Areas:

1. Query processing stages
 2. Query optimization
 3. BI tools overview
 4. Reporting and visualization
 5. Examples of open-source BI tools
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Topic 7: OLAP Application using KNIME

Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation
-

PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
 2. Each member must participate
 3. Assign roles:
 1. Presenter
 2. Designer (PPT)
 3. Researcher
 4. Demo/Case handler
-

◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
 1. Definition and importance
 2. Core Concepts
 1. Explanation with diagrams
 3. Process/Architecture
 1. Step-by-step explanation
 4. Real-World Example
 1. Case study or application
 5. Tool/Demo (if applicable)
 1. KNIME or BI tool
 6. Conclusion
 1. Summary + future scope
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◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
 1. Diagrams
 2. Flowcharts

3. 3. Tables (for comparison)
 3. Avoid:
 1. Too much text
 2. Copy-paste content
-

◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
 1. Eye contact
 2. Clear voice
 3. Logical flow



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES



OLAP and Data Cubes

(Real-Time Business Analytics Using Amazon Sales Data)

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Presented by:
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CSA1603
(Data Warehousing and Data Mining)

❖ Definition

OLAP (Online Analytical Processing) is a technology used to analyze large amounts of business data from different perspectives. It helps managers generate reports, identify trends, and make faster business decisions.

❖ Why do businesses need OLAP?

- Analyze millions of records quickly
- Compare sales across multiple dimensions
- Generate interactive reports
- Support strategic decision-making



FIG - 01 : OLAP Dashboard for Amazon Sales Analysis

Example: Amazon analyzes customer purchases to understand buying patterns and improve business performance.

DATA CUBES

➤ Definition

A Data Cube organizes data into multiple dimensions, making analysis faster and easier.

➤ **Common Dimensions :** Product , Location , Time

➤ Example :

Product	City	Month	Sales
Laptop	Chennai	January	8 Cr
Mobile	Bengaluru	January	12 Cr
Tablet	Hyderabad	February	5 Cr

➤ Key Idea:

A Data Cube allows users to analyze data from different angles without scanning millions of records.



FIG - 02 : OLAP Data Cube
Product ,Time , Place

FOUR IMPORTANT OLAP OPERATIONS

01

Slice

View data for one dimension.

Example: Sales only in Chennai.

03

Drill Down

Move from summary to detailed data.

India → Tamil Nadu → Chennai

02

Dice

Select a specific combination of data.

Example: Mobile sales in Chennai during January.

04

Roll Up

Summarize detailed data.
City → State → Country



REAL-TIME EXAMPLE (AMAZON)

- ❑ **How Amazon Uses OLAP :** Every day Amazon processes millions of transactions.
- ❑ **Managers need answers like:**
 - Which products are selling the most?
 - Which cities generate the highest revenue?
 - Which month has the maximum sales?
 - Which warehouse requires more stock?

Using OLAP and Data Cubes, these answers are available within seconds.

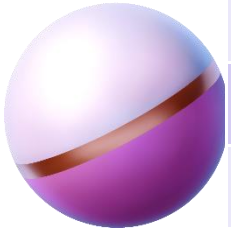
❑ **Benefits**

- ✓ Faster reporting
- ✓ Increased profit
- ✓ Improved customer satisfaction
- ✓ Better inventory planning

Amazon Sales Dashboard



❖ Sample Sales Report



City	Smart Phone Sales
Chennai	12 Cr
Bengaluru	9 Cr
Mumbai	10 Cr
Hyderabad	8 Cr

❖ Business Insight

Chennai has the highest smartphone sales.

❖ Decision

Increase inventory in Chennai before the next sale.

INTERACTIVE ACTIVITY

Let's Think Like a Business Analyst!

❖ QUESTION - 01

- ❖ Amazon wants to view only Chennai sales.

❖ Which OLAP operation should be used?

- ☐ A. Roll Up
- ☐ B. Slice
- ☐ C. Dice
- ☐ D. Drill Down

✓ Correct Answer: Slice

Reason:

Slice selects data from one specific dimension (Chennai).

❖ QUESTION - 02

- ❖ Amazon wants to analyze sales from India → Tamil Nadu → Chennai

❖ Which OLAP operation should be used?

- ☐ A. Slice
- ☐ B. Roll Up
- ☐ C. Dice
- ☐ D. Drill Down

✓ Correct Answer: Drill Down

Reason:

Drill Down allows users to move from summary-level data to more detailed data.



CONCLUSION



- **Key Takeaways**

- ❖ OLAP enables fast analysis of large datasets.
- ❖ Data Cubes organize information across multiple dimensions.
- ❖ Businesses use OLAP for faster and better decision-making.
- ❖ Amazon uses OLAP to analyze customer behavior, optimize inventory, and improve sales performance.

- **Future Scope**

- ☐ Artificial Intelligence (AI)
- ☐ Predictive Analytics
- ☐ Cloud Data Warehousing
- ☐ Real-Time Business Intelligence



Engineer to Excel



Thank You